

Literature-Implied Status Note for arXiv:2004.00891

Trace-Norm Consistency Without Reflexivity

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Status

literature_implied_answer (full trace-class conclusion). This is an identification of a known 1972 theorem, not a new theorem.

Source question

Mollenhauer–Klus–Schütte–Koltai, *Kernel Autocovariance Operators of Stationary Processes: Estimation and Convergence*, arXiv:2004.00891v2, Remark 11 on PDF page 9, observes that the trace class $S_1(\mathcal{H})$ is nonreflexive when $\mathcal{H}$ is infinite-dimensional. It asks whether reflexivity is necessary for the almost-sure norm convergence in the Banach-valued Birkhoff theorem, with convergence of the empirical kernel autocovariance in trace norm as the motivating unresolved case.

Supporting literature

Pop-Stojanovic, *On ergodic theorem for a Banach valued random sequence*, J. Austral. Math. Soc. **13** (1972), 501–507, Corollary 2 on journal page 505 (proof on page 506; PDF pages 5–6), states strong almost-sure convergence of the averages of a Bochner-integrable strictly stationary Banach-valued sequence to its invariant conditional expectation. No reflexivity assumption is imposed. In an ergodic system the invariant conditional expectation is constant and equals the Bochner expectation.

Identification

Fix an integer lag η and set

$$F_t = \varphi(X_{t+\eta}) \otimes \varphi(X_t) \in S_1(\mathcal{H}).$$

The rank-one trace-norm identity and Cauchy–Schwarz give

$$\mathbb{E}\|F_0\|_{S_1} = \mathbb{E}[\|\varphi(X_\eta)\| \|\varphi(X_0)\|] \leq (\mathbb{E}\|\varphi(X_\eta)\|^2)^{1/2} (\mathbb{E}\|\varphi(X_0)\|^2)^{1/2} < \infty.$$

Under the source paper’s measurability assumptions, (F_t) is therefore a Bochner-integrable strictly stationary $S_1(\mathcal{H})$ -valued sequence. Pop-Stojanovic’s Corollary 2 applies, and ergodicity identifies the limit as $\mathbb{E}F_0 = C(\eta)$. Consequently

$$\|C_n(\eta) - C(\eta)\|_{S_1(\mathcal{H})} \longrightarrow 0 \quad \text{almost surely.}$$

This remains true when $\mathcal{H}$ is infinite-dimensional, so reflexivity is not necessary for the convergence in question. Moreover, $\|A\|_{S_p} \leq \|A\|_{S_1}$ for every $1 \leq p \leq \infty$, so the same statement yields all Schatten-norm conclusions for each fixed lag.

Provenance and scope

The relation is agent-identified: the 1972 supporting paper predates and does not discuss kernel autocovariance operators. It nevertheless gives the exact Banach-valued ergodic theorem needed by Remark 11. The conclusion uses the source paper’s stationarity, ergodicity, strong measurability, and second-moment assumptions. Integer lags form a countable set, so a common probability-one event for all such lags follows by countable intersection if desired.

References

- [1] M. Mollenhauer, S. Klus, C. Schütte, and P. Koltai, *Kernel Autocovariance Operators of Stationary Processes: Estimation and Convergence*, arXiv:2004.00891v2 (2022).
- [2] Z. R. Pop-Stojanovic, *On ergodic theorem for a Banach valued random sequence*, J. Austral. Math. Soc. **13** (1972), 501–507, doi:10.1017/S1446788700009253.